

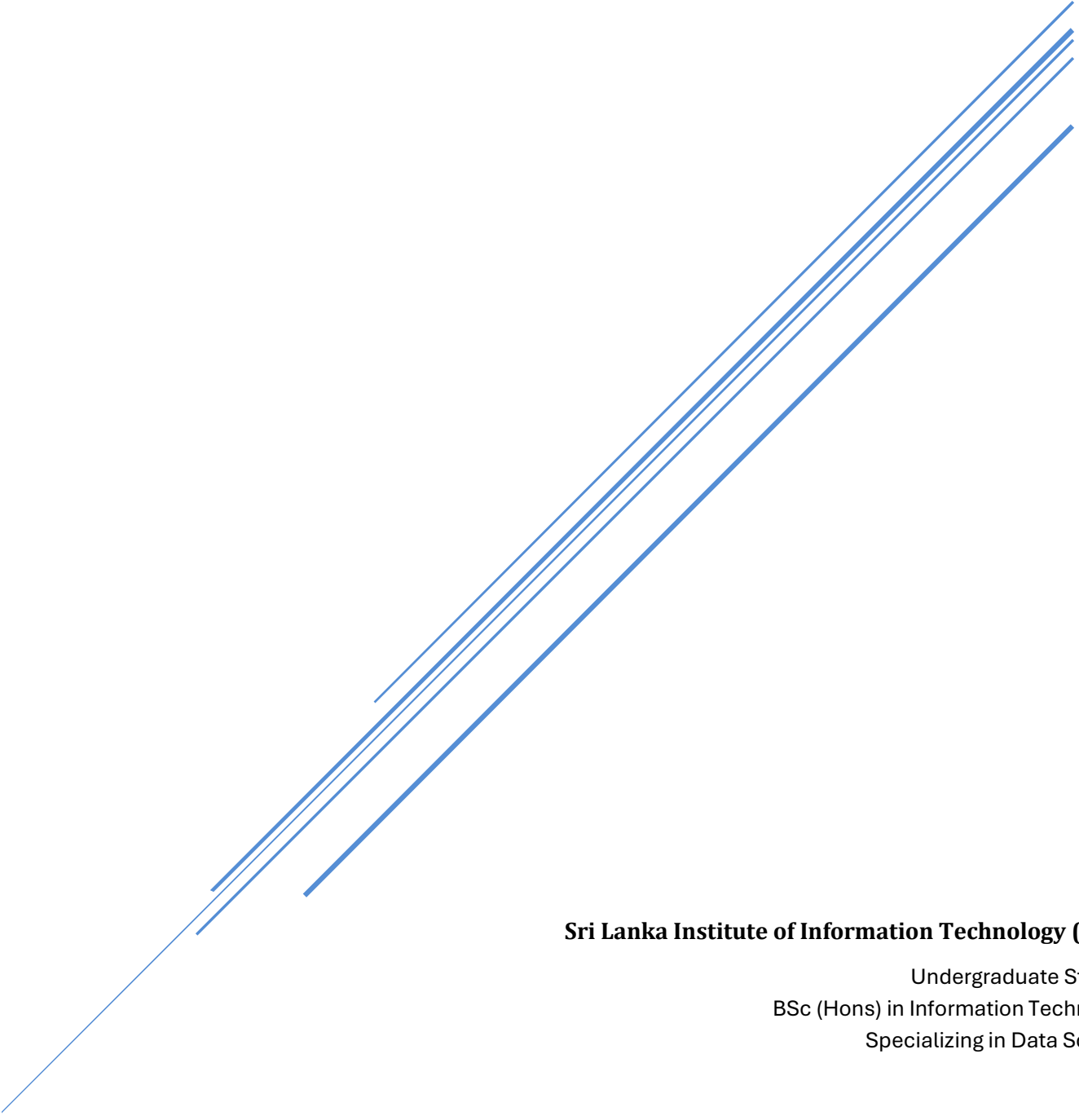
Customer Service Process Analytics Using R

Validating the Relationship Between Smooth Service Processes and Customer Experience

A Data Analytics Project

Prepared by

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1. Abstract

This report validates the analytical statement, 'Customers who experience smooth service processes report better overall experiences', using an airline customer satisfaction dataset. The dataset contains 129,880 customer records and 23 original variables related to passenger demographics, travel characteristics, service ratings, flight delays, and satisfaction status. The analysis was conducted using R Studio and followed three stages: descriptive analytics, inferential analytics, and predictive analytics. A Smooth Service Process Score was created from variables such as ease of online booking, online boarding, check-in service, gate location, baggage handling, and departure/arrival time convenience. An Overall Experience Score was created from service experience variables such as seat comfort, inflight entertainment, online support, onboard service, leg-room service, cleanliness, and food and drink. Descriptive results showed that customers in the high smoothness group had the highest average overall experience score and satisfaction rate. Inferential results confirmed a statistically significant relationship between smooth process level and overall customer experience. Predictive analytics using logistic regression showed strong classification performance, with an AUC close to 0.91. Overall, the findings support the selected analytical statement, while the results should be interpreted as evidence of association rather than definite causation.

2. Introduction

Customer experience is a key concern in service industries because customers evaluate service providers based on the quality, convenience, reliability, and comfort of the complete service process. In airline services, the customer journey includes several stages such as online booking, check-in, boarding, baggage handling, flight punctuality, onboard comfort, and post-flight experience. If these stages operate smoothly, customers are more likely to report higher satisfaction and a better overall experience.

This project uses statistical modelling principles to validate whether smoother service processes are associated with better customer experiences. The selected topic is suitable for a Data Science assignment because it can be tested using measurable service rating variables, hypothesis testing, correlation analysis, and predictive modelling.

3. Problem Statement and Objectives

Problem Statement: The purpose of this study is to determine whether customers who experience smooth service processes report better overall experiences.

The main research question is: Do customers with higher smooth service process scores report higher overall experience scores and higher satisfaction levels?

The objectives of the analysis are to:

- Create meaningful analytical variables representing service process smoothness and overall experience.
- Use descriptive analytics to summarize satisfaction patterns and compare customer groups.
- Use inferential analytics to test whether differences between smooth-process groups are statistically significant.
- Use predictive analytics to model customer satisfaction and assess whether service-process variables help predict satisfaction.

4. Background

In service analytics, customer satisfaction is commonly treated as an outcome influenced by service quality, process efficiency, reliability, and customer perceptions. For airline customers, smooth processes include easy online booking, convenient timing, efficient check-in, reliable baggage handling, and smooth boarding. These process-related factors can directly affect how customers evaluate the service provider.

The selected dataset is appropriate because it includes both service process variables and a final satisfaction outcome. This allows the statement to be tested using descriptive summaries, hypothesis testing, correlation analysis, and logistic regression.

5. Methodology

The analysis followed a structured statistical modelling workflow using R Studio. The workflow included data import, cleaning, feature construction, descriptive analysis, inferential testing, predictive modelling, and interpretation.

Stage	Methods Used
Data preparation	Missing value handling, duplicate checking, encoding, feature creation, outlier review
Descriptive analytics	Summary statistics, grouped means, distributions, bar charts, box plots, heatmap
Inferential analytics	One-way ANOVA, chi-square test, Pearson correlation, two-sample t-test, confidence intervals
Predictive analytics	Logistic regression, 80:20 train-test split, confusion matrix, accuracy, precision, recall, F1-score, ROC-AUC

6. Data Description and Preparation

The dataset used in this project was the Airline Customer Satisfaction dataset. It contains 129,880 records and 23 original variables. The variables include customer type, age, travel type, travel class, flight distance, service ratings, delay information, and final satisfaction status.

The dataset contained 393 missing values in the arrival_delay_in_minutes variable, representing approximately 0.3% of records. These values were replaced with 0 because the records did not contain a recorded arrival delay. This decision retained all customer records and allowed arrival delay to be categorized as no delay where applicable. Duplicate checking found 0 duplicate rows.

Preparation Step	Action Taken
Column cleaning	Column names were converted into R-friendly snake_case names.
Missing values	393 missing arrival delay values were replaced with 0.
Duplicate records	0 duplicate rows were found.
Smooth Process Score	Mean score was calculated using six service-process variables.
Overall Experience Score	Mean score was calculated using eight customer experience variables.
Delay treatment	Arrival delay was grouped into no, minor, moderate, and major delay.
Outlier treatment	Extreme delay values were capped at the 99th percentile for modelling.

7. Descriptive Analytics

Descriptive analytics was used to understand the distribution of satisfaction, the level of service-process smoothness, and the relationship between smoothness groups and overall customer experience.

Metric	Value
Total customers	129,880
Satisfied customers	71,087
Dissatisfied customers	58,793
Satisfaction rate	54.7%
Mean smooth process score	3.307
Mean overall experience score	3.312
Mean age	39.43
Mean flight distance	1981.41

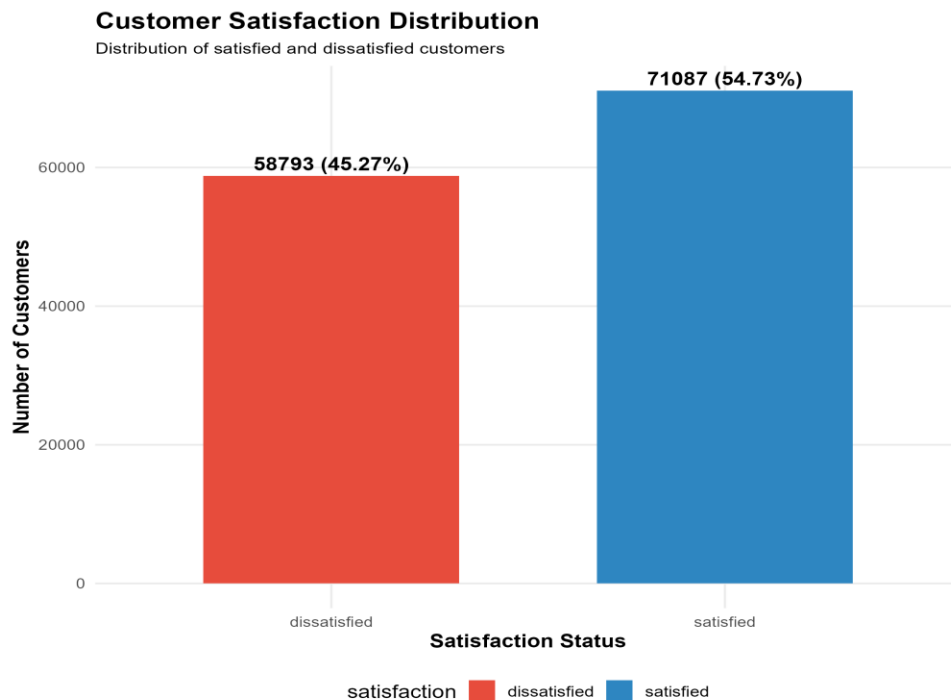


Figure 1. Distribution of customer satisfaction status.

The dataset contains more satisfied customers than dissatisfied customers. Satisfied customers represent approximately 54.7% of the records, while dissatisfied customers represent approximately 45.3%.

Smooth Process Group	Customer Count	Mean Smooth Score	Mean Overall Experience	Satisfaction Rate
Low Smoothness	15,302	2.0672	2.4441	21.5%
Moderate Smoothness	56,051	2.9761	3.0612	46.8%
High Smoothness	58,527	3.9481	3.7801	71.0%

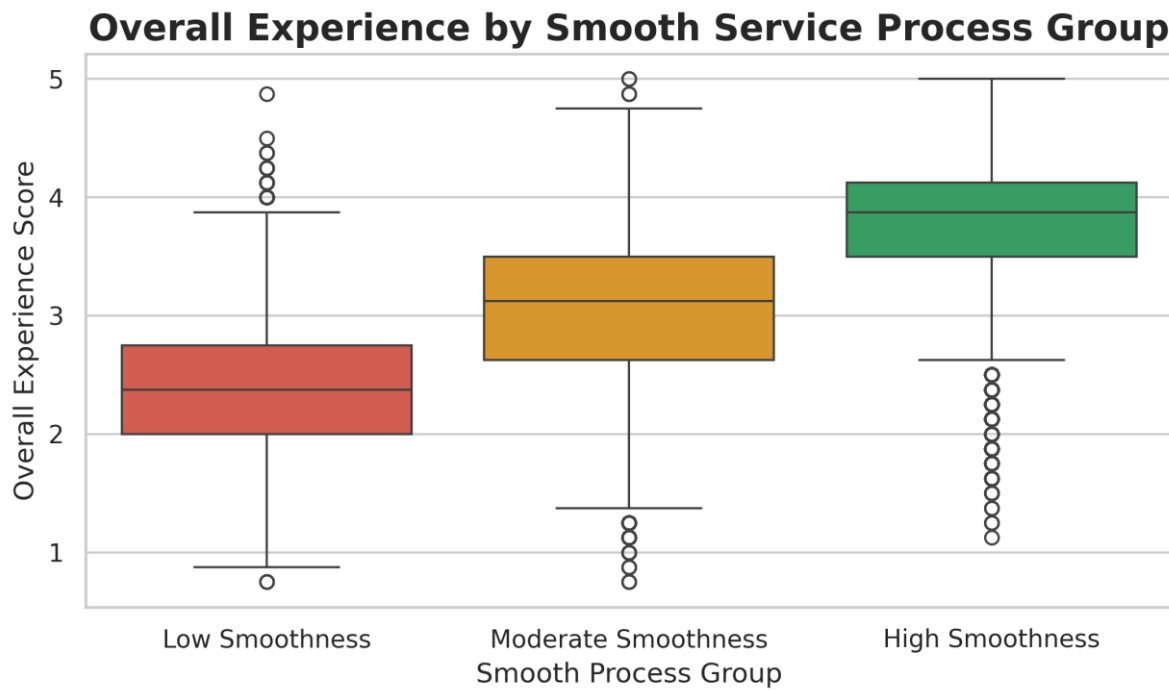


Figure 2. Overall experience score by smooth service process group.

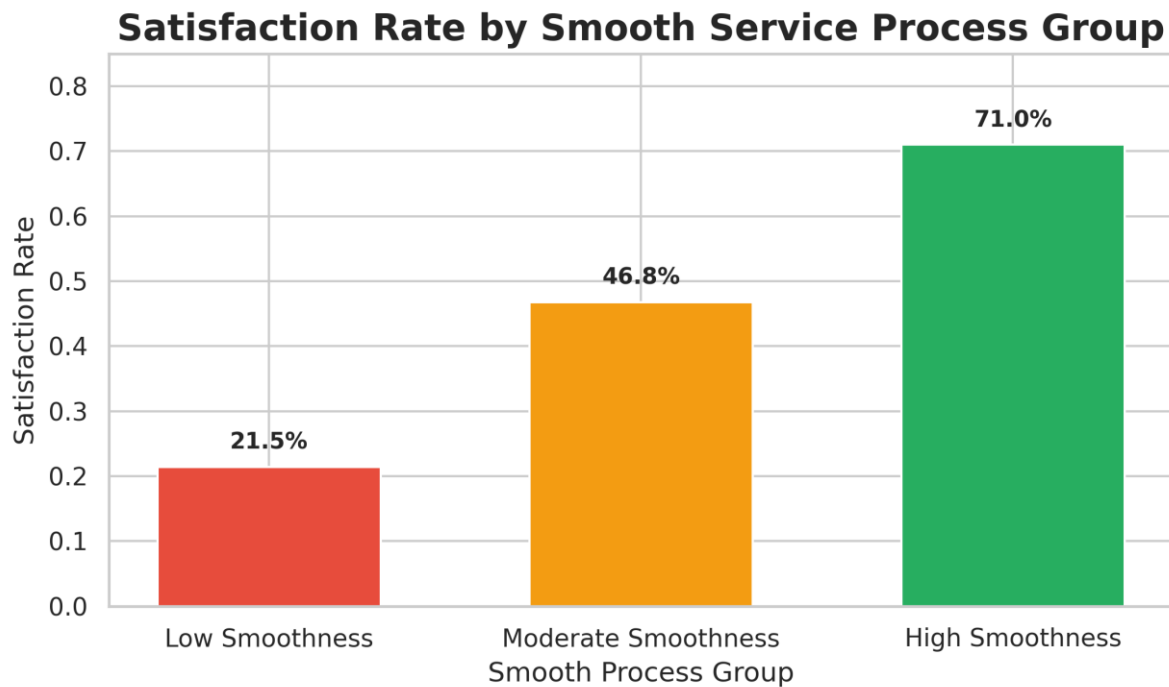


Figure 3. Satisfaction rate by smooth service process group.

The descriptive results show a clear increasing pattern. Customers in the low smoothness group had the lowest mean overall experience score and satisfaction rate. Customers in the high smoothness group had the highest mean overall experience score and satisfaction rate. This provides initial descriptive support for the selected analytical statement.

Arrival Delay Group	Customer Count	Mean Overall Experience	Satisfaction Rate
No Delay	73,146	3.3473	58.6%
Minor Delay	27,176	3.2994	53.0%
Moderate Delay	20,470	3.2559	47.9%
Major Delay	9,088	3.198	43.8%

Arrival delay also showed a practical relationship with satisfaction. Customers with no arrival delay had the highest satisfaction rate, while customers with major delays had the lowest satisfaction rate. This supports the idea that process disruptions can reduce the customer experience.

Correlation Heatmap of Service Process and Experience Variables

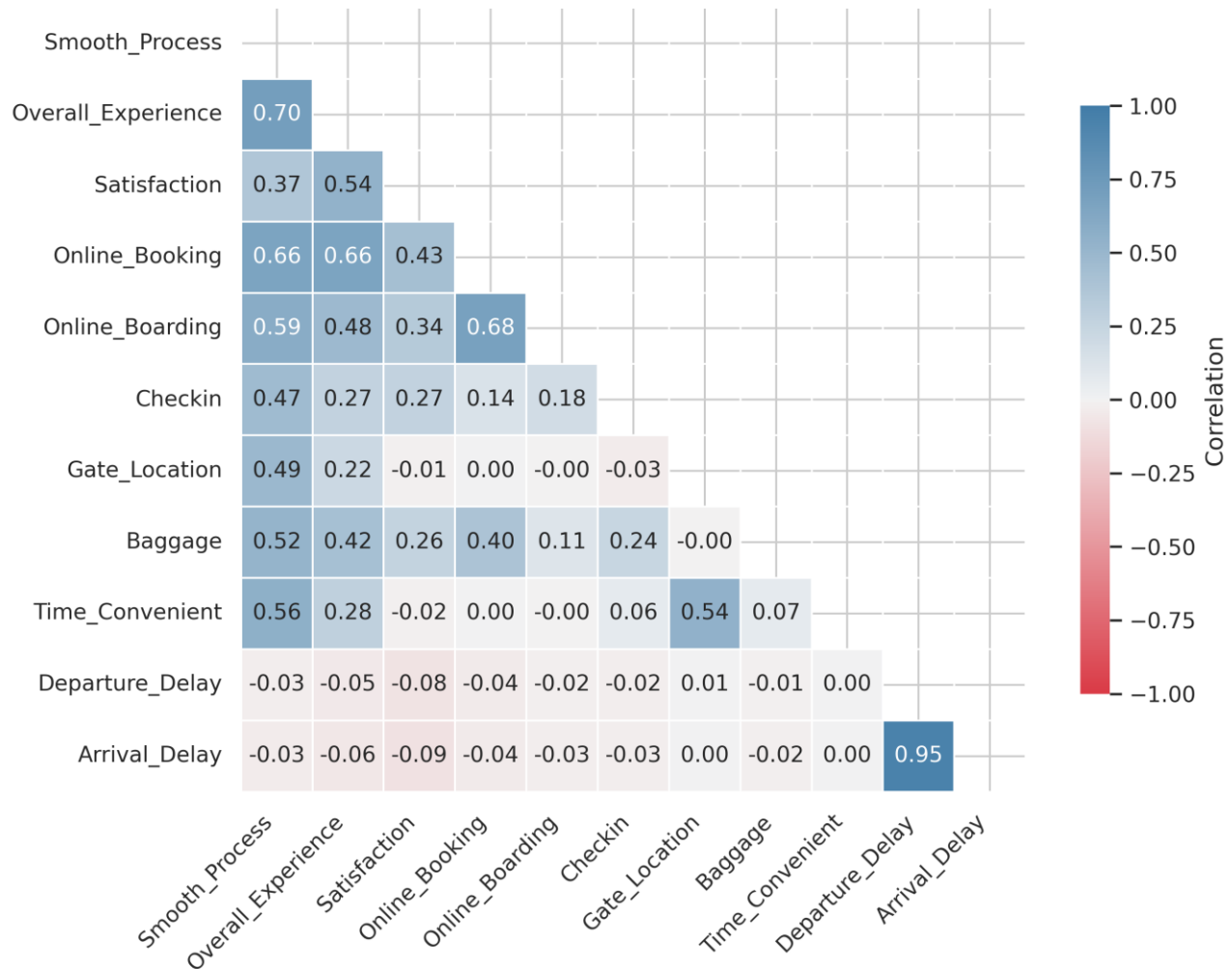


Figure 4. Correlation heatmap for service process and experience variables.

The correlation analysis showed a strong positive relationship between smooth_process_score and overall_experience_score, with $r = 0.705$. This indicates that customers with smoother service process ratings also tended to report better overall experience scores.

8. Inferential Analytics

Inferential analytics was conducted to determine whether the observed descriptive differences were statistically significant. A 5% significance level was used for hypothesis testing.

8.1 One-Way ANOVA

The one-way ANOVA tested whether mean overall experience scores differed across low, moderate, and high smooth process groups.

H0: There is no significant difference in mean overall experience scores among the smooth process groups.

H1: At least one smooth process group has a significantly different mean overall experience score.

Test	F Statistic	p-value	Decision	Interpretation
One-way ANOVA	42909.348	< 0.001	Reject H0	Overall experience differs significantly across smooth process groups.

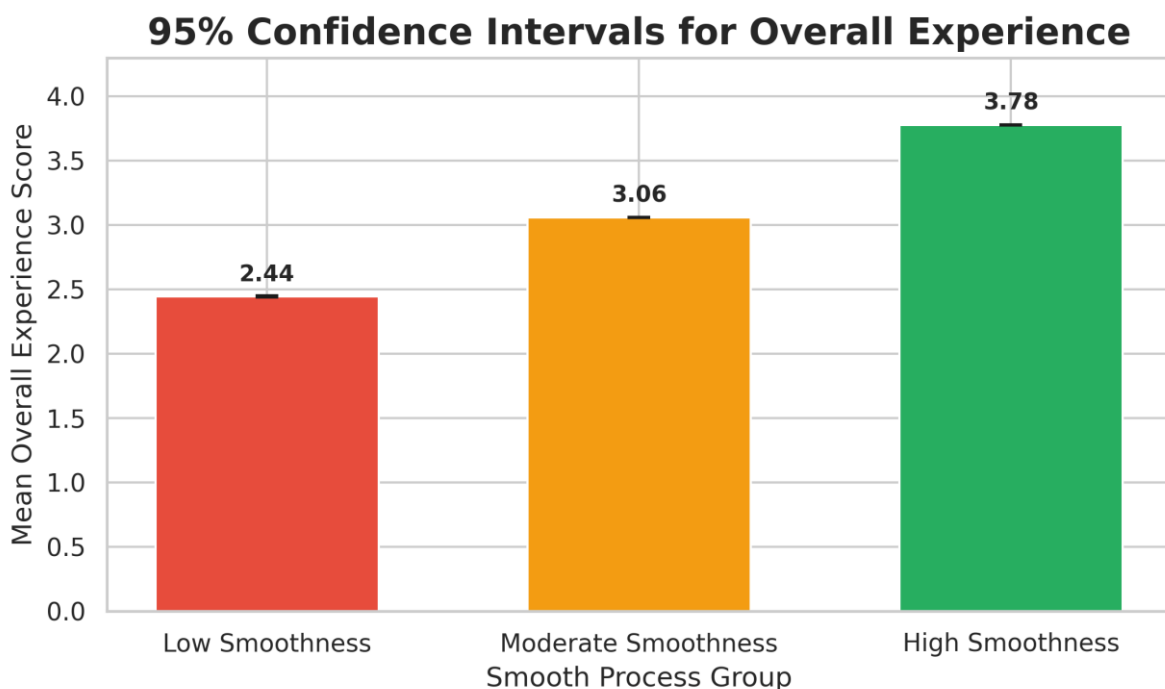


Figure 5. Mean overall experience with 95% confidence intervals.

The ANOVA result was statistically significant. Therefore, the null hypothesis was rejected. The confidence intervals also show a clear increase in mean overall experience from low to high smoothness groups.

8.2 Chi-Square Test of Association

The chi-square test was used to determine whether smooth process group and satisfaction status are associated.

H0: Smooth process group and satisfaction status are independent.

H1: Smooth process group and satisfaction status are associated.

Test	Statistic	df	p-value	Decision	Interpretation
Chi-square test	14539.626	2	< 0.001	Reject H0	Smooth process group and satisfaction status are significantly associated.

8.3 Correlation and t-test Results

Analysis	Statistic	p-value	Main Result
Pearson correlation	$r = 0.705$	< 0.001	Smooth process score and overall experience score have a strong positive relationship.
Two-sample t-test	$t = -143.476$	< 0.001	Satisfied customers had a higher mean smooth process score (3.550) than dissatisfied customers (3.014).

The two-sample t-test showed that satisfied customers had a significantly higher average smooth process score than dissatisfied customers. The 95% confidence interval for the difference between dissatisfied and satisfied customers was $[-0.543, -0.528]$, confirming that the mean score for dissatisfied customers was lower.

9. Predictive Analytics

Predictive analytics was used to determine whether customer satisfaction could be predicted using service-process and customer-experience variables. Since the target variable was binary, satisfied or dissatisfied, logistic regression was selected as the appropriate model.

The dataset was split into 80% training data and 20% testing data. The training set was used to build the model, and the testing set was used to evaluate performance on unseen data.

Metric	Value
Training records	103,904
Testing records	25,976
Accuracy	0.829
Precision	0.842
Recall	0.847
F1-score	0.844
ROC-AUC	0.903



Figure 6. ROC curve for the logistic regression model.

ROC Curve Comparison

Full Model AUC = 0.908 | Selected Model AUC = 0.908

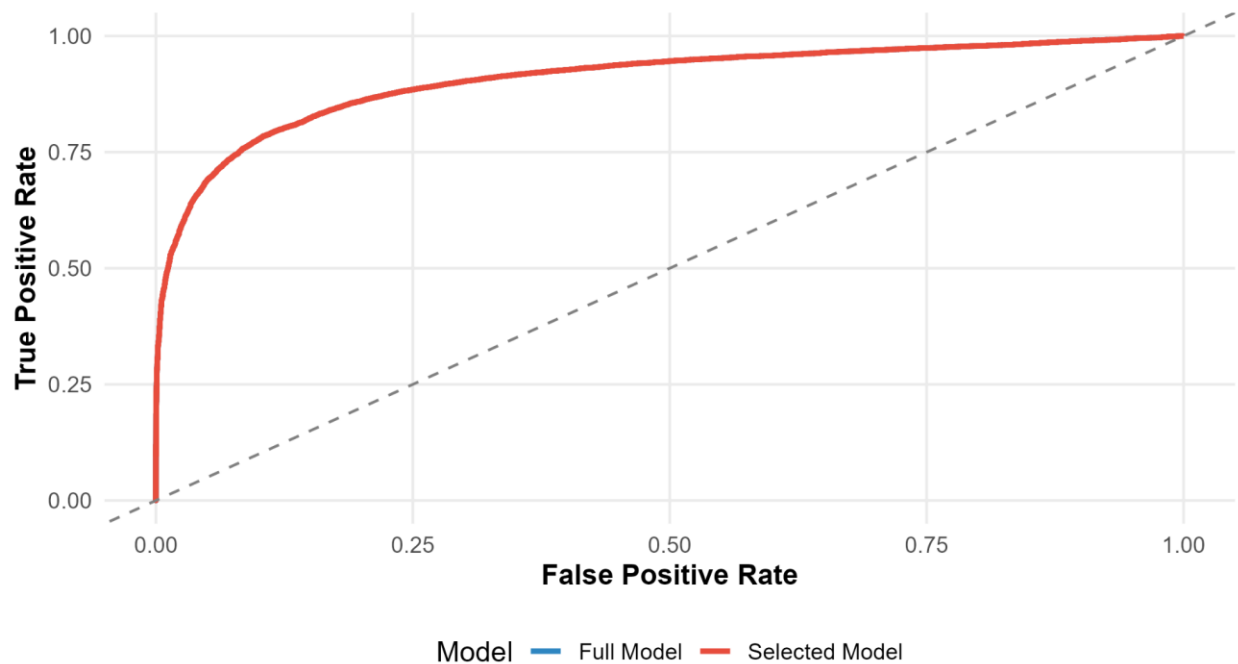


Figure 7. Full model and selected model ROC comparison from the project outputs.

The logistic regression model achieved strong predictive performance. The project output showed that both the full model and selected model achieved an AUC of approximately 0.908, meaning the selected model preserved predictive performance while improving interpretability. This supports the usefulness of service-process and experience variables for predicting customer satisfaction.

Actual Class	Predicted Not Satisfied	Predicted Satisfied
Actual Not Satisfied	9,496	2,263
Actual Satisfied	2,178	12,039

10. Discussion

The combined results provide consistent evidence supporting the analytical statement. Descriptive analytics showed that high smoothness customers had higher satisfaction rates and higher overall experience scores. Inferential analytics confirmed that these differences were statistically significant. Predictive analytics showed that customer satisfaction can be predicted with strong performance using service-process and experience-related variables.

The strongest evidence came from the smooth process group comparison. The high smoothness group had a mean overall experience score of 3.780 and a satisfaction rate of 71.0%. In contrast, the low smoothness group had a mean overall experience score of 2.444 and a satisfaction rate of 21.5%. This difference is both statistically significant and practically meaningful.

The findings also show the importance of reducing service disruptions. Customers with no arrival delay had a satisfaction rate of 58.6%, while customers with major delays had a satisfaction rate of 43.8%. Although delays are only one part of the service process, they clearly influence the overall experience.

An important academic consideration is that the analysis identifies association, not definite causation. While smoother service processes are strongly related to better customer experience, other factors such as travel class, customer type, flight distance, and onboard service conditions may also influence satisfaction.

11. Conclusion

Based on descriptive, inferential, and predictive analytics, the statement “Customers who experience smooth service processes report better overall experiences” is supported by the analyzed dataset.

Descriptive analytics showed a clear increase in satisfaction and overall experience from low to high smoothness groups. Inferential analytics confirmed statistically significant differences and associations, with p-values below 0.001. Predictive analytics showed that logistic regression could classify customer satisfaction with strong performance, supported by an ROC-AUC close to 0.91.

Therefore, the final conclusion is that smooth service processes are strongly associated with better customer experience and higher satisfaction in the airline service context.

12. Limitations

- The dataset is limited to airline customer satisfaction and may not represent all service industries.
- The satisfaction ratings are survey-based and may contain response bias.
- The analysis uses secondary data, so the researcher has no control over how the data was originally collected.
- Correlation and regression results show association, not direct causation.
- Some constructed scores, such as `smooth_process_score` and `overall_experience_score`, are based on selected variables and may not capture every aspect of customer experience.

13. Recommendations

- Service providers should improve digital service stages such as online booking and online boarding because these are part of the smooth service process.
- Airlines should reduce delays and improve delay communication because customers with higher delay levels showed lower satisfaction.
- Operational teams should focus on check-in, baggage handling, and gate-location convenience to improve the complete customer journey.
- Management should monitor smooth service process scores as a key performance indicator for customer experience.

14. References

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- R Core Team. (n.d.). R: A language and environment for statistical computing. R Foundation for Statistical Computing.